

L'Hôpital's Rule: Statement and Proof

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1 Prerequisite

Rolle's theorem is taken as known and is not reproved here.

Theorem 1.1 (Rolle). *Let $h : [p, q] \rightarrow \mathbb{R}$ be continuous on $[p, q]$ and differentiable on (p, q) , with $h(p) = h(q)$. Then there exists $\xi \in (p, q)$ such that $h'(\xi) = 0$.*

2 Cauchy's Mean Value Theorem

Lemma 2.1 (Cauchy). *Let $f, g : [p, q] \rightarrow \mathbb{R}$ be continuous on $[p, q]$ and differentiable on (p, q) . Then there exists $\xi \in (p, q)$ such that*

$$[f(q) - f(p)] g'(\xi) = [g(q) - g(p)] f'(\xi).$$

If in addition $g'(x) \neq 0$ for every $x \in (p, q)$, then $g(p) \neq g(q)$ and

$$\frac{f(q) - f(p)}{g(q) - g(p)} = \frac{f'(\xi)}{g'(\xi)}.$$

Proof. Define $h : [p, q] \rightarrow \mathbb{R}$ by

$$h(x) = [f(q) - f(p)] g(x) - [g(q) - g(p)] f(x).$$

As a linear combination of f and g , h is continuous on $[p, q]$ and differentiable on (p, q) . A direct computation gives

$$h(p) = f(q)g(p) - g(q)f(p), \quad h(q) = f(q)g(p) - g(q)f(p),$$

so $h(p) = h(q)$. By Rolle's theorem there is $\xi \in (p, q)$ with $h'(\xi) = 0$; since

$$h'(x) = [f(q) - f(p)] g'(x) - [g(q) - g(p)] f'(x),$$

this is exactly the claimed identity.

Now suppose $g'(x) \neq 0$ throughout (p, q) . If $g(p) = g(q)$, Rolle's theorem applied to g itself on $[p, q]$ would produce a point of (p, q) at which g' vanishes, contradicting the hypothesis. Hence $g(p) \neq g(q)$, and since also $g'(\xi) \neq 0$ (else the identity above would force $g(p) = g(q)$ as well, again a contradiction), dividing by $g'(\xi)[g(q) - g(p)]$ yields the second formula. $\square$

3 L'Hôpital's Rule

Theorem 3.1 (L'Hôpital). *Let $-\infty \leq c < d \leq +\infty$, and let $f, g : (c, d) \rightarrow \mathbb{R}$ be differentiable, with $g'(x) \neq 0$ for every $x \in (c, d)$. Suppose*

$$\lim_{x \rightarrow c^+} \frac{f'(x)}{g'(x)} = A, \quad A \in \mathbb{R} \cup \{-\infty, +\infty\},$$

(when $c = -\infty$, “ $x \rightarrow c^+$ ” is read simply as $x \rightarrow -\infty$), and suppose in addition that either

- (a) $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^+} g(x) = 0$, or
 (b) $\lim_{x \rightarrow c^+} g(x) = +\infty$ (or $-\infty$).

Then

$$\lim_{x \rightarrow c^+} \frac{f(x)}{g(x)} = A.$$

Proof. Step 0: g is injective on (c, d) . If $g(x) = g(y)$ for some $x \neq y$ in (c, d) , Rolle's theorem applied to g on the interval between x and y would produce a zero of g' in (c, d) , contradicting the hypothesis $g' \neq 0$. So g takes each value at most once on (c, d) ; being also continuous, it is strictly monotonic there (a standard consequence of the intermediate value theorem).

Step 1: the case $A \in \mathbb{R}$. Fix $\varepsilon > 0$ and choose real numbers q, r with

$$A - \varepsilon < q < A < r < A + \varepsilon.$$

Since $f'/g' \rightarrow A$ as $x \rightarrow c^+$, there is a point $b \in (c, d)$ such that

$$q < \frac{f'(t)}{g'(t)} < r \quad \text{for all } t \in (c, b). \quad (1)$$

For any x, y with $c < x < y < b$, Step 0 gives $g(x) \neq g(y)$, so Cauchy's theorem (Lemma 1) applies on $[x, y]$: there exists $\xi = \xi(x, y) \in (x, y) \subset (c, b)$ with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(\xi)}{g'(\xi)}.$$

Combined with (1), this gives

$$q < \frac{f(x) - f(y)}{g(x) - g(y)} < r \quad \text{for all } c < x < y < b. \quad (2)$$

Case (a). Fix $y \in (c, b)$. By Step 0, g is strictly monotonic on (c, d) with $\lim_{x \rightarrow c^+} g(x) = 0$, so 0 is not attained on (c, d) : hence $g(y) \neq 0$. Letting $x \rightarrow c^+$ in (2), and using $f(x) \rightarrow 0$, $g(x) \rightarrow 0$, the middle term tends to $f(y)/g(y)$, and weak inequalities survive the limit:

$$q \leq \frac{f(y)}{g(y)} \leq r \quad \text{for every } y \in (c, b). \quad (3)$$

Case (b). Fix $y \in (c, b)$. Because $g(x) \rightarrow +\infty$ as $x \rightarrow c^+$ (the case $g \rightarrow -\infty$ is identical after replacing g by $-g$), we may choose x close enough to c that $g(x) > 0$ and $g(x) > g(y)$. Multiplying (2) by the positive quantity $[g(x) - g(y)]/g(x)$ gives

$$q \left(1 - \frac{g(y)}{g(x)}\right) < \frac{f(x)}{g(x)} - \frac{f(y)}{g(x)} < r \left(1 - \frac{g(y)}{g(x)}\right). \quad (4)$$

Let $x \rightarrow c^+$ with y fixed. Since $g(x) \rightarrow +\infty$, both $g(y)/g(x) \rightarrow 0$ and $f(y)/g(x) \rightarrow 0$, so (4) yields

$$q \leq \liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq \limsup_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq r. \quad (5)$$

In either case, (3) or (5) gives

$$q \leq \liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq \limsup_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq r.$$

This holds for *every* admissible pair q, r with $A - \varepsilon < q < A < r < A + \varepsilon$. Letting $\varepsilon \rightarrow 0^+$ forces $q \rightarrow A, r \rightarrow A$, hence

$$\liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} = \limsup_{x \rightarrow c^+} \frac{f(x)}{g(x)} = A,$$

i.e., $\lim_{x \rightarrow c^+} f(x)/g(x) = A$.

Step 2: the case $A = +\infty$ (the case $A = -\infty$ is symmetric). Fix any $r \in \mathbb{R}$. Since $f'/g' \rightarrow +\infty$, there is $b \in (c, d)$ with $f'(t)/g'(t) > r$ for all $t \in (c, b)$. Repeating the argument of Step 1 using only this one-sided bound (there is no companion q to track) produces, in either case (a) or (b),

$$\liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} \geq r.$$

Since $r \in \mathbb{R}$ was arbitrary, $\liminf_{x \rightarrow c^+} f(x)/g(x) = +\infty$, i.e., $\lim_{x \rightarrow c^+} f(x)/g(x) = +\infty$. $\square$

4 Extensions

Corollary 4.1 (Left-hand and two-sided limits). *The theorem holds equally for $x \rightarrow c^-$: if f, g are differentiable on (e, c) for some $e < c$, with $g' \neq 0$ there, and $\lim_{x \rightarrow c^-} f'(x)/g'(x) = A$ together with the analogous hypothesis (a) or (b), then $\lim_{x \rightarrow c^-} f(x)/g(x) = A$. Indeed, set $\tilde{f}(x) = f(2c - x)$ and $\tilde{g}(x) = g(2c - x)$ on $(c, 2c - e)$; as $x \rightarrow c^+$, $2c - x \rightarrow c^-$, and $\tilde{f}'(x)/\tilde{g}'(x) = f'(2c - x)/g'(2c - x) \rightarrow A$, so Theorem 2 applies to $\tilde{f}, \tilde{g}$ and yields the claim after undoing the substitution. Consequently, if f, g are differentiable on a full punctured neighborhood $(e, c) \cup (c, d)$ of a finite point c and both one-sided derivative-ratio limits equal A , then $\lim_{x \rightarrow c} f(x)/g(x) = A$.*

Corollary 4.2 (Limits at infinity). *The case $x \rightarrow +\infty$ reduces to Theorem 2 via $t = 1/x$: if $F(t) = f(1/t)$, $G(t) = g(1/t)$ for t near 0^+ , the chain rule gives*

$$\frac{F'(t)}{G'(t)} = \frac{f'(1/t)}{g'(1/t)},$$

so $\lim_{x \rightarrow +\infty} f'(x)/g'(x) = A$ implies $\lim_{t \rightarrow 0^+} F'(t)/G'(t) = A$; hypotheses (a)/(b) transfer identically, so Theorem 2 gives $\lim_{t \rightarrow 0^+} F(t)/G(t) = A$, which is $\lim_{x \rightarrow +\infty} f(x)/g(x) = A$. The case $x \rightarrow -\infty$ is analogous via $t = -1/x$.

Remark 4.3. The hypothesis that $\lim_{x \rightarrow c^+} f'(x)/g'(x)$ exists cannot be dropped, and the theorem is not reversible. Take $f(x) = x + \sin x$, $g(x) = x$ on $(0, \infty)$; both diverge to $+\infty$ as $x \rightarrow \infty$, and

$$\frac{f(x)}{g(x)} = 1 + \frac{\sin x}{x} \rightarrow 1,$$

so the quotient converges. Yet

$$\frac{f'(x)}{g'(x)} = 1 + \cos x$$

oscillates between 0 and 2 and has no limit as $x \rightarrow \infty$. This is not a contradiction: the theorem states a *sufficient*, not necessary, condition for evaluating $\lim f/g$.